Software Requirements Specification

Gym Membership Management System

Part 2 of 9 • Requirements Engineering

Project	Gym Membership Management System	**Version**	1.0
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1 Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Gym Membership Management System. It describes what the system must do and how well it must perform, so that the team can design, build, and test it against clear and agreed requirements.

1.2 Scope

The system is a desktop application that helps a small or medium-sized gym manage its members, membership plans, subscriptions, payments, and attendance, and produce simple reports. It follows the scope defined in the Part 1 proposal. Online payment, a mobile app, biometric or card-reader hardware, and workout or diet planning are outside this version.

1.3 Definitions

Term	Meaning
SRS	Software Requirements Specification, this document.
Subscription	The period during which a member may use the gym, based on a chosen membership plan.
Membership plan	A package with a name, duration (monthly, quarterly, or yearly), and price.
Check-in	A record that a member has entered the gym on a given date and time.
FR / NFR	Functional Requirement / Non-Functional Requirement.

2 User Roles

Role	Description	Main permissions
Administrator	The gym owner or manager who controls the system.	Everything Staff can do, plus manage staff accounts and membership plans, correct payments, and view all reports.
Staff (Receptionist)	Front-desk employee who handles daily work.	Register and search members, renew subscriptions, record payments, record check-ins, and view basic reports.
Member	A person who subscribes to the gym. Members do not log in to the system.	Receive their subscription status, receipts, and payment history through staff.

3 User Requirements

User requirements describe, in the users' own words, what each user needs to achieve. They are made detailed in Section 4.

ID	Who	User requirement	Detailed in
UR-01	Administrator	I want to control who can use the system and what they can do.	FR-01 to FR-04
UR-02	Administrator	I want to set up membership plans and their prices.	FR-10
UR-03	Staff	I want to register a new member quickly and find an existing member easily.	FR-05 to FR-09
UR-04	Staff	I want to start or renew a subscription and see when it ends.	FR-11 to FR-14
UR-05	Staff	I want to record a payment and give the member a receipt.	FR-15 to FR-18
UR-06	Staff	I want to record when a member comes to the gym.	FR-19 to FR-21
UR-07	Administrator	I want reports on members, income, and expiring subscriptions to make decisions.	FR-22 to FR-26
UR-08	Member	I want to know my subscription status and see my payment history.	FR-13, FR-17

4 System Requirements

System requirements are the detailed requirements the software must meet. Together, the **functional requirements** (this section) and the **non-functional requirements** (Section 5) make up the system requirements.

> **Functional requirements** describe **what** the system must do. Each is stated as "The system shall...". Priority: **High** = needed for the core prototype; **Medium** = important but can be done last.

4.1 Login and User Management

ID	Requirement	Priority
FR-01	The system shall require a username and password before allowing access.	High
FR-02	The system shall assign each user one role (Administrator or Staff) and only show the functions allowed for that role.	High
FR-03	The system shall allow the Administrator to add, edit, and deactivate staff accounts.	High
FR-04	The system shall allow a logged-in user to log out.	Medium

4.2 Member Management

ID	Requirement	Priority
FR-05	The system shall register a new member with name, phone number, gender, date of birth, and join date, and give the member a unique member ID.	High
FR-06	The system shall allow authorized users to edit a member's information.	High
FR-07	The system shall allow searching for members by name, member ID, or phone number.	High
FR-08	The system shall allow a member to be deactivated or reactivated instead of deleted, so that history is kept.	Medium
FR-09	The system shall warn the user if a member with the same phone number already exists.	Medium

4.3 Plans and Subscriptions

ID	Requirement	Priority
FR-10	The system shall allow the Administrator to add, edit, and disable membership plans (name, duration, price).	High
FR-11	The system shall assign a plan to a member and calculate the subscription end date automatically from the start date and plan duration.	High
FR-12	The system shall allow a subscription to be renewed for an existing member.	High
FR-13	The system shall show each member's subscription status as Active, Expiring Soon, or Expired.	High
FR-14	The system shall list subscriptions that will expire within a set number of days (default 7).	Medium

4.4 Payments

ID	Requirement	Priority
FR-15	The system shall record a payment with the member, amount, date, payment method, and related subscription.	High
FR-16	The system shall generate a unique receipt number for every payment and allow the receipt to be displayed or printed.	High
FR-17	The system shall show the full payment history of a member.	High
FR-18	The system shall allow only the Administrator to correct or cancel a recorded payment, and shall log the change.	Medium

4.5 Attendance

ID	Requirement	Priority
FR-19	The system shall record a member's check-in with the date and time.	High
FR-20	The system shall warn the user when a member with an expired subscription tries to check in.	Medium
FR-21	The system shall show attendance history by member and by date.	Medium

4.6 Reports

ID	Requirement	Priority
FR-22	The system shall produce a report of active members.	High
FR-23	The system shall produce a report of expired members.	High
FR-24	The system shall produce a monthly income report based on recorded payments.	High
FR-25	The system shall produce a report of subscriptions that are about to expire.	Medium
FR-26	The system shall allow reports to be printed.	Medium

5 Non-Functional Requirements

Non-functional requirements describe **how well** the system must work: its quality, speed, security, and ease of use.

ID	Category	Requirement
NFR-01	Usability	A new staff member shall be able to register a member and record a payment after about 30 minutes of training.
NFR-02	Usability	The interface shall use clear labels and show understandable messages for errors and confirmations.
NFR-03	Performance	Searching for a member shall return results within 2 seconds for up to 2,000 members.
NFR-04	Performance	Any report shall be generated within 5 seconds.
NFR-05	Reliability	Data shall be saved immediately, so no information is lost if the program is closed.
NFR-06	Reliability	The system shall handle invalid input without crashing and shall provide a way to back up the database.
NFR-07	Security	Passwords shall be stored in hashed form, never as plain text.
NFR-08	Security	Access to functions and data shall be restricted by user role, and login shall be blocked temporarily after 5 failed attempts.
NFR-09	Data integrity	The system shall validate input (for example valid dates, positive amounts, required fields) and keep member IDs unique.
NFR-10	Maintainability	The code shall be organized into separate parts (interface, logic, data) with comments so that it is easy to modify.
NFR-11	Compatibility	The system shall run on Windows 10 or later on a standard office computer.

6 System Constraints

- The project must be completed by a team of three students within the semester schedule.
- The system is a desktop application for a single gym location and works without an internet connection.
- Only free or student-licensed tools may be used; no paid services. The final technology choice is made in Part 4 (Architecture).
- Online payment, mobile app, biometric or card-reader hardware, and workout or diet planning are excluded (see Part 1 scope).
- The prototype is tested with sample data only, not real members' personal data.
- The user interface language is English.

7 Assumptions

- The gym has at least one Windows computer at the reception desk.
- Staff have basic computer skills.
- Members do not log in; staff give them their subscription status and payment history on request.
- Each member has at most one active subscription at a time, and phone numbers identify members uniquely.
- Payments are made at the gym (for example in cash) and the system only records them, using one currency.
- Membership plans have fixed durations (monthly, quarterly, yearly) and prices set by the Administrator.

8 Link to Project Objectives

Each objective from the Part 1 proposal is covered by the requirements below.

Part 1 objective	Covered by
1. Store member information in one searchable database	FR-05 to FR-09, NFR-03
2. Manage plans and track subscription start and expiry dates	FR-10 to FR-14
3. Record payments and reduce errors	FR-15 to FR-18, NFR-09
4. Record member check-in attendance	FR-19 to FR-21
5. Generate simple reports	FR-22 to FR-26, NFR-04